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Vellore Institute of Technology

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AGE AND GENDER DETECTION USING CAFFE MODEL

A PROJECT REPORT

for

AARTIFICIAL INTELLIGENCE (SWE4010)

in

Integrated M.Tech(SE)

by

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FALL SEMESTER, 2023-24

Under the Guidance of

Prof. CHIRANJI LAL CHOWDHARY

Associate Professor, SCORE

School of Computer Science Engineering and Information Sciences

December, 2023

DECLARATION BY THE CANDIDATE

We here by declare that the project report entitled “**AGE AND GENDER DETECTION USING CAFFE MODEL**” submitted by us to Vellore Institute of Technology University, Vellore in partial fulfilment of the requirement for the award of the course **Artificial Intelligence (SWE4010)** is a record of bonafide project work carried out by us under the guidance of **Prof. Chiranji Lal Chowdhary**. We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other course.

Place : Vellore

Signature

Date : 21-11-23

Abstract

Deep Learning has found huge applications in the field of Computer vision. Some of the most important applications of computer vision are in the fields that deal with facial data. Face Detection and recognition are being widely used in security-based applications. In this project, we are going to talk about two of the most important applications of face-based deep learning, age and gender prediction from a facial image. Deep learning has emerged as a powerful and versatile tool with extensive applications in the field of computer vision. Among these applications, one of the most significant and pervasive is in the domain of facial data analysis. Face detection and recognition have seen widespread use, particularly in security-based applications. This project delves into two pivotal facets of face-based deep learning: age and gender prediction from facial images. By harnessing the capabilities of deep learning models, we explore the exciting potential to infer not only the age but also the gender of an individual from a single facial image. This work is not only technically intriguing but also holds practical implications in various domains, from personalized marketing and content recommendation to surveillance, public safety, and beyond. In this project, we delve into the techniques, challenges, and ethical considerations surrounding age and gender prediction, offering insights into the remarkable synergy of deep learning and computer vision.



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CERTIFICATE

This is to certify that the project report entitled “**AGE AND GENDER DETECTION USING CAFFE MODEL**” submitted by **VADHINI JHAVER (20MIS0022), MURAHARI CHERMINI (20MIS0042), SREEJA VADANALA (20MIS0139), LAKSHMI G MENON (20MIS0346), VARSHA JHAVER (21MIS0164), MANISHA T (21MIS0192)** to Vellore Institute of Technology University, Vellore in partial fulfillment of the requirement for the award of the course **Artificial Intelligence (SWE4010)** is a record of bonafide work carried out by them under my guidance.

Guide:

Prof. Chiranjilal Chowdhary

I. INTRODUCTION

1. Overview

Deep Learning has found huge applications in the field of Computer vision. Some of the most important applications of computer vision are in the fields that deal with facial data. Face Detection and recognition are being widely used in security-based applications.

In this guided project, we are going to talk about two of the most important applications of face-based deep learning, age, and gender prediction from a facial image.

To build this project we will be using an OpenCV and Caffe model where the weights will be taken from pre-trained models each for facial detection, age, and gender. The prediction will result on a real-time basis as well as on a manual basis.

2. Purpose

By the end of this project, we will:

1. Know fundamental concepts and techniques of Neural Network.
2. Will gain a broad understanding of image data.
3. Know how to draw bounding boxes according to your requirement.
4. Know how to build a web application using the Flask framework.

II. PRELIMINARIES

In the preliminary stages of developing an age and gender detection system using deep learning, the key steps include data collection with diverse and labeled datasets, data preprocessing to prepare the data for model training, model selection, and architecture design, hyper parameter tuning, ethical considerations to ensure privacy and fairness, deployment planning, continuous monitoring, and updates to maintain accuracy over time, and documentation for user understanding and legal compliance, while keeping a strong focus on ethical implications and potential biases in model predictions.

III. LITERATURE SURVEY

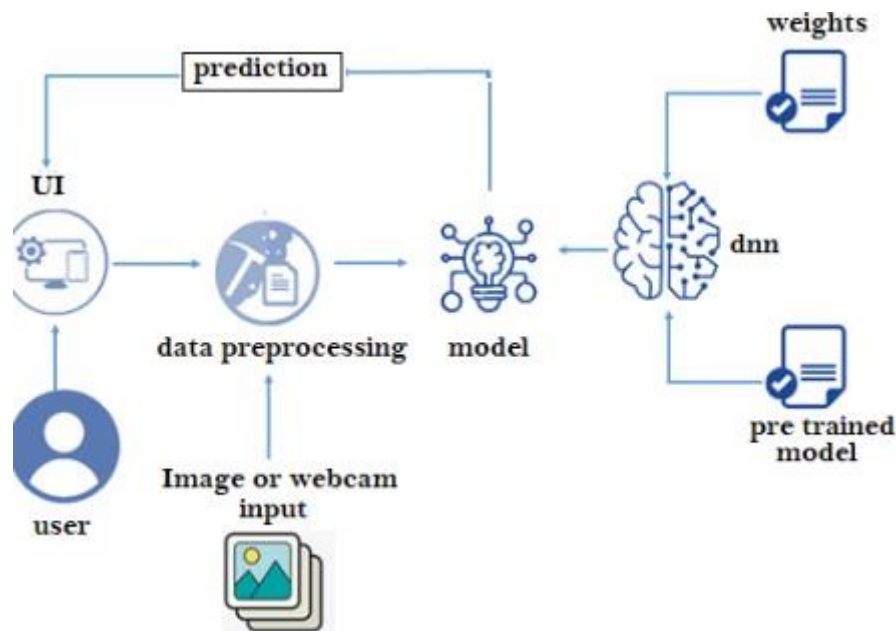
A literature survey for age and gender detection in deep learning involves reviewing existing research, papers, and studies related to the field. Below is a summary of some notable work in this area up to my knowledge cutoff date in January 2022:

1. "Deep Convolutional Neural Networks for Age and Gender Classification" by Gil Levi and Tal Hassner (2015): This influential paper introduced a popular approach for age and gender estimation using deep CNNs, achieving impressive accuracy on large-scale datasets.
2. "Age and Gender Classification using Convolutional Neural Networks" by Rothe, Rother, and Denzler (2016): This work presented a deep CNN model trained for age and gender classification and explored different techniques for handling age labels.
3. "Deep Expectation of Real and Apparent Age from a Single Image Without Facial Landmarks" by Yi Li et al. (2017): This paper proposed a method for age estimation that does not rely on facial landmarks, making it robust to a wide range of images.
4. "Age and Gender Estimation from Facial Images Using Convolutional Neural Networks" (2017) by G. Antipov et al. This research paper delves into age and gender estimation from facial images using CNNs, offering insights into the state-of-the-art methodologies at the time.
5. "Age and Gender Classification with Deep Learning" by Zhifei Zhang et al. (2019): This research explored deep learning techniques for age and gender classification and introduced the concept of "latent factors" for age prediction.
6. "Deep Learning-Based Gender Classification Using Histograms of Oriented Gradients" by Sungwook Kim et al. (2020): This study combined deep learning with traditional feature extraction techniques to improve gender classification accuracy.
7. "Deep Learning-Based Age Estimation and Gender Classification in Unconstrained Environments" by Balakrishna Chennupati et al. (2021): This paper discussed age and gender classification in uncontrolled environments, focusing on addressing challenges such as pose variation and image quality.
8. "FairFace: Face Attribute Dataset and Analysis" by K. Karkkainen et al. (2020): This work introduced the FairFace dataset, which includes age and gender annotations, and highlighted the importance of fairness and bias mitigation in age and gender classification.

9. "Deep Learning for Age Estimation: A Survey, a Taxonomy, and an Objective Evaluation" (2017) by M. Rothe et al. This survey paper focuses on age estimation using deep learning techniques and could contain valuable insights into age detection methodologies.

IV. PROPOSED ALGORITHM WITH EXPLANATION

The proposed system integrates OpenCV and Caffe models to enhance face-based deep learning applications. By utilising pre-trained models for facial detection, age, and gender, the system aims to achieve real-time predictions. The pipeline involves capturing facial data in real-time, processing it using the pre-trained models, and providing instantaneous results for age and gender. The proposed system is expected to offer improved accuracy and efficiency compared to traditional methods.



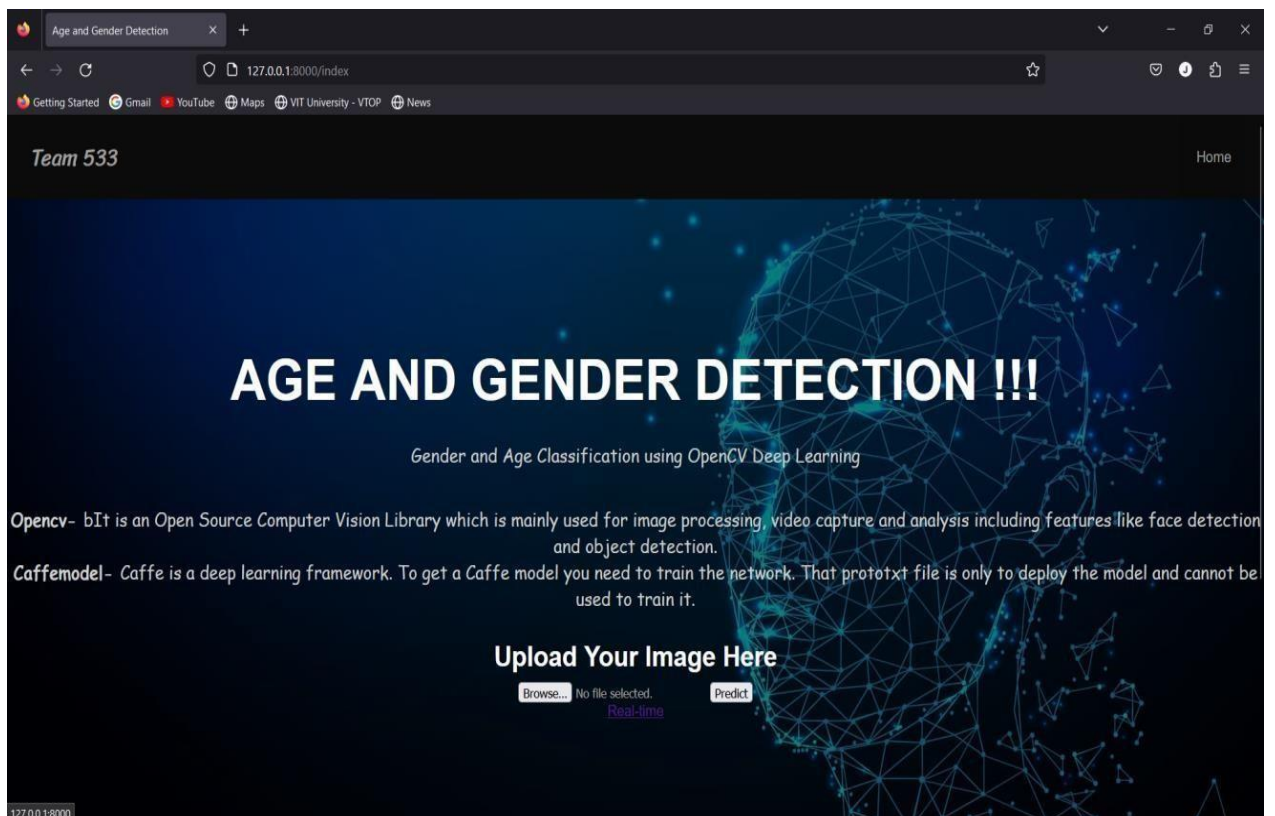
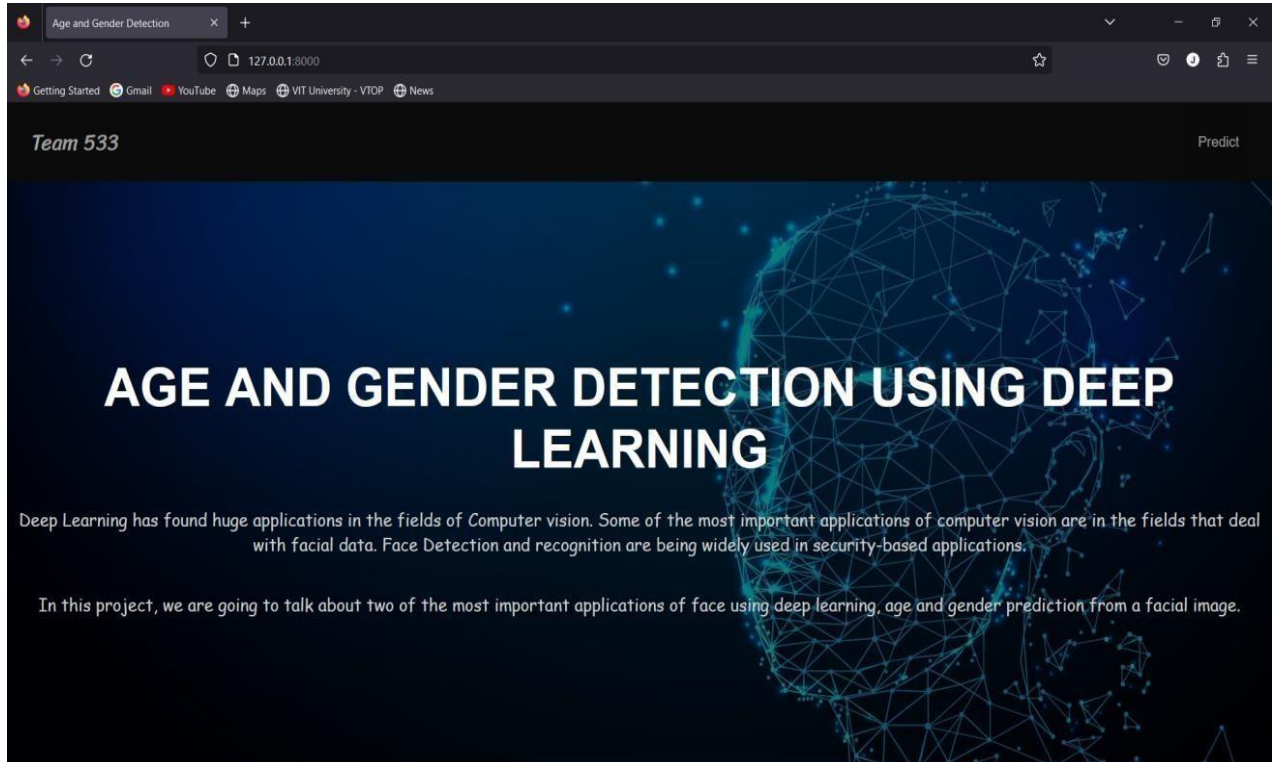
Caffe models:

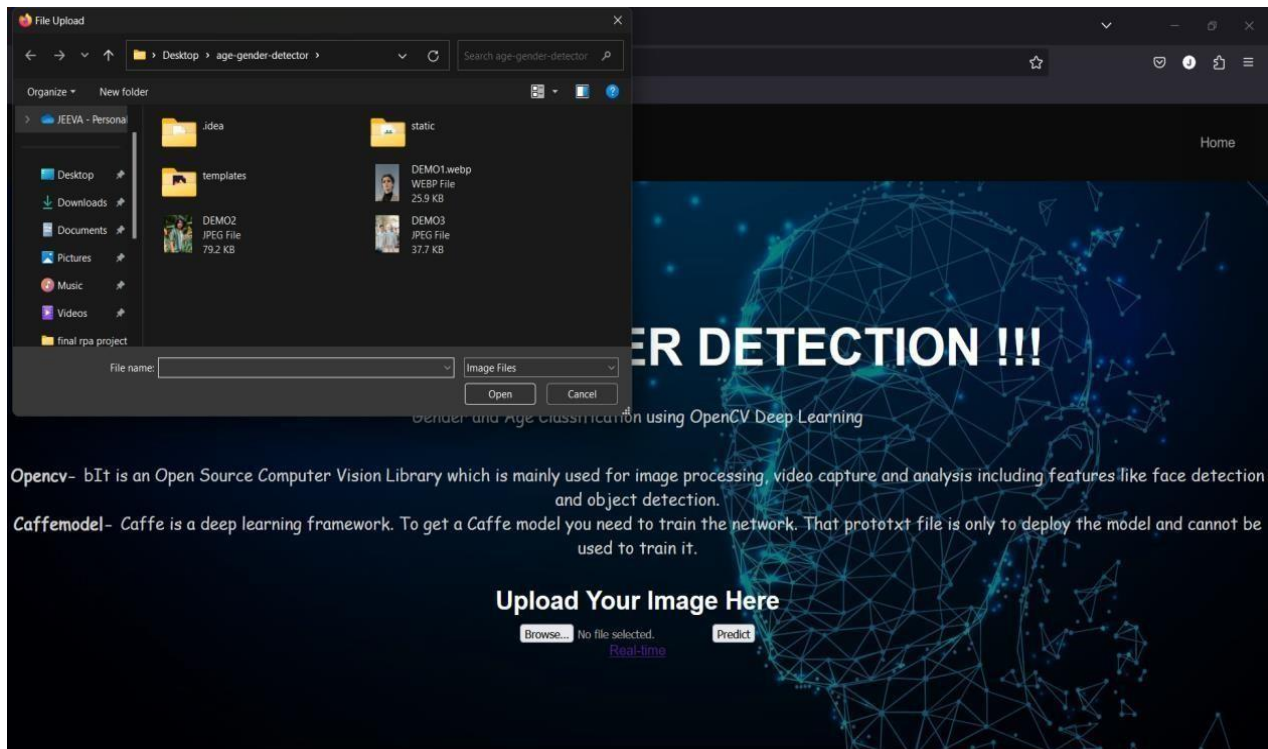
1. Caffe (Convolutional Architecture for Fast Feature Embedding) is a deep learning framework developed by the Berkeley Vision and Learning Center (BVLC) that is particularly well-suited for CNNs.
2. Caffe uses its own file format for model weights and architectures, which is referred to as "caffemodel" files.
3. Predefined Caffe models, or caffemodels, are pre-trained models that have already been trained on large datasets for specific computer vision tasks. These models can be used for fine-tuning on custom datasets or for feature extraction.

V. EXPERIMENTS RESULTS

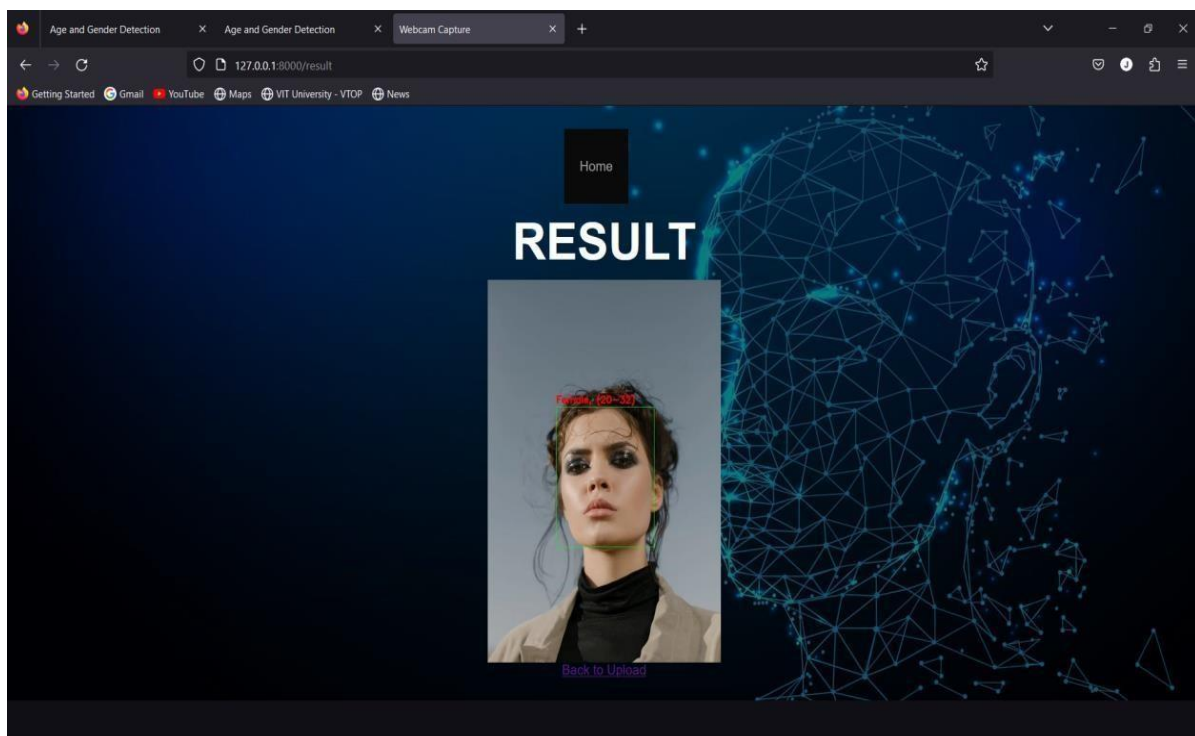
Some of the screenshots that define the result of our project:

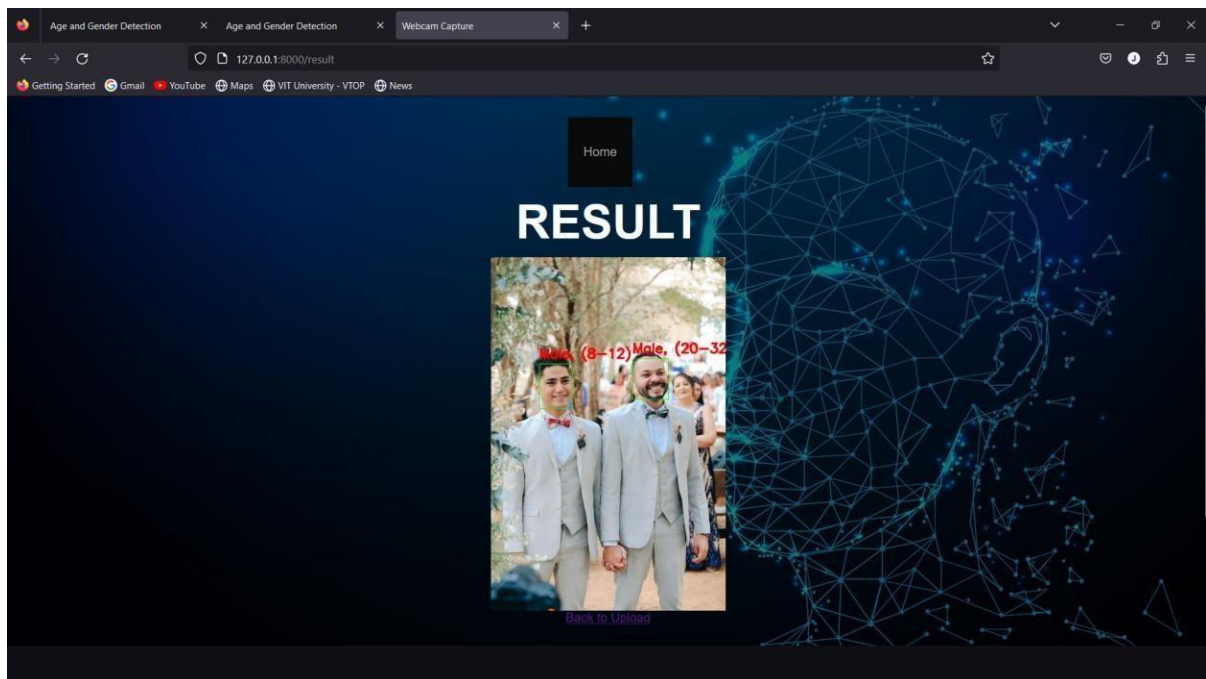
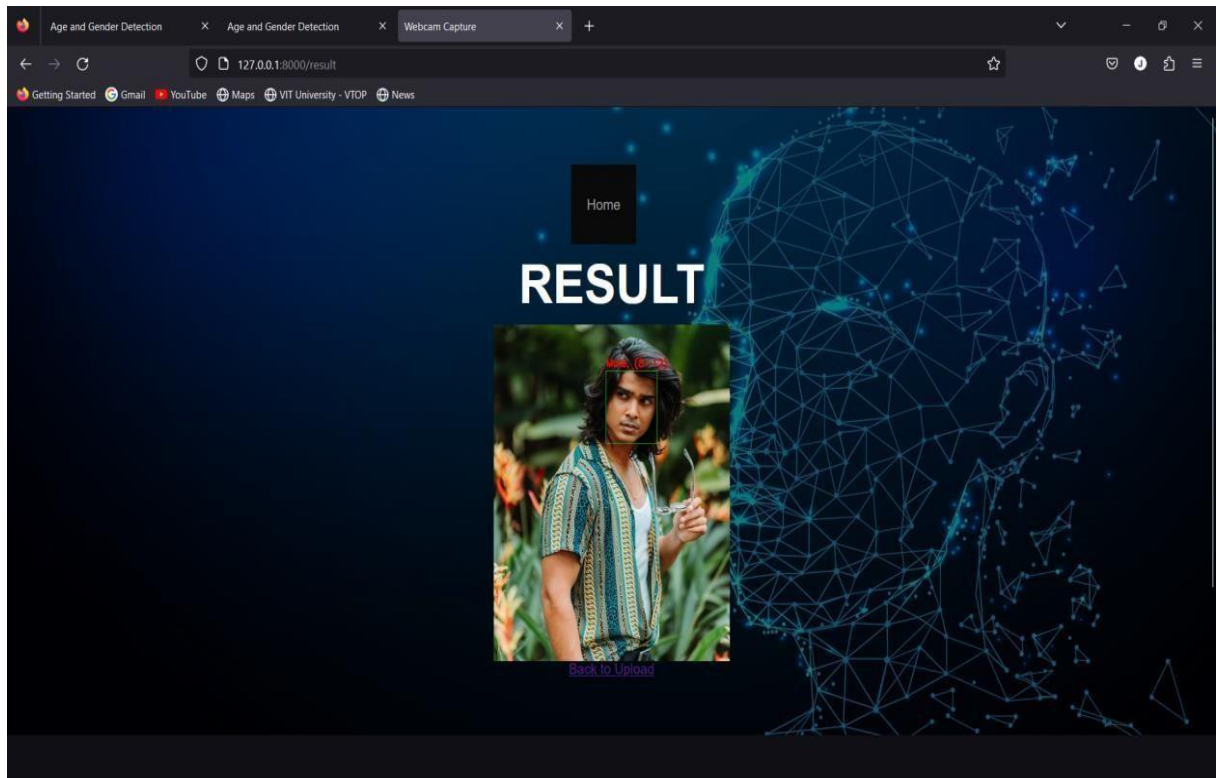
Website:



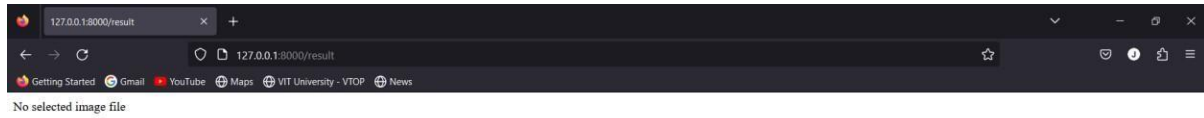


Output (uploaded images):





Output (when face cannot be detected or no image is selected):



VI. COMPARATIVE STUDY

Customisation:

CNN algorithms are highly customisable, making them suitable for a wide range of applications.

Predefined Caffe models have fixed architectures and are less customisable in terms of changing the network structure. You can fine-tune them for specific tasks but within certain constraints.

Training Effort:

CNN algorithms require training from scratch, which can be computationally expensive and time-consuming.

Predefined Caffe models come with pre-trained weights, saving you the effort and resources required for training.

Applicability:

CNN algorithms are versatile and can be used for building custom models for specific tasks.

Predefined Caffe models are excellent for tasks similar to those they were pre-trained for and are especially useful when you have limited labeled data for your task.

VII. CONCLUSION AND FUTURE WORK

1. CONCLUSION

Age detection and gender detection ai have many different uses. For example, in-store and front-of-shop age detection and gender detection are very effective even in retail businesses. By using these technologies, businesses can identify customer segments, change or develop marketing strategies according to this segment, and change their product range.

For example, some software can detect the age range of users passing by a business and suggest advertisements that will affect this age range. It is possible to use age detection and gender detection not only for sales purposes but also for security purposes because algorithms can make more accurate age and gender determinations than humans.

The only negative aspect of these algorithms is that they have extremely complex work principles, as mentioned above, and they also require large data sets. This means a loss of both time and money. If you cannot find specialists in the field, it is challenging to learn all these technologies. In addition, software developed in this regard can be highly costly.

2. FUTURE SCOPE

Since, the only negative aspect of these algorithms is that they have extremely complex work principles, as mentioned above, and they also require large data sets. This means a loss of both time and money. If you cannot find specialists in the field, it is challenging to learn all these technologies. In addition, software developed in this regard can be highly costly.

Hence, the solution for this disadvantage can be considered as an advantage in future. If this is done, it can be expanded in vast culture and can help people in many ways.

REFERENCES

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